

class11

Kate Ruiz (PID: A17671200)

Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~ 250 thousand structures (we saw this in the last lab). The main protein sequence database has over **200 million** sequences! Only 0.125 % of sequences have a known structure - this is called the “structure knowledge gap”.

```
(250000 / 200000000) * 100
```

```
[1] 0.125
```

Structures are much harder to determine than sequences. They are expensive. On average ~ \$1 million each. They take on average 3-5 years to solve.

EBI AlphaFold Database

The EBI has a database of pre-computed Alpha Fold (AF) models called AFDB. This is growing all the time and can be useful to check before running AF ourselves.

Running AlphaFold

We can download and run locally (on our own computers) but we need a GPU. Or we can use “cloud” computing to run this on someone else’s computer :-)

We will use ColabFold < <https://github.com/sokrypton/ColabFold> >

We previously found there was no AFDB entry for our HIV sequence

```
>HIV-Pr-Dimer
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQ
```

Here we will use AlphaFold2_mmseqs2

8. Custom analysis of resulting models

```
library(bio3d)
library(pheatmap)
library(jsonlite)
results_dir <- "hivpr_23119/"
pdb_files <- list.files(path=results_dir,
  pattern="*.pdb",
  full.names = TRUE)
# Print our PDB file names
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)
# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1 name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2 name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3 name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4 name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5 name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3

pdbs

```

1                                     .               .               .               .               50
[Truncated_Name:1]hivpr_2311      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_2311      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_2311      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_2311      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_2311      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
*****
1                                     .               .               .               .               50

51                                     .               .               .               .               100
[Truncated_Name:1]hivpr_2311      GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_2311      GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_2311      GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_2311      GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_2311      GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
51                                     .               .               .               .               100

101                                    .               .               .               .               150
[Truncated_Name:1]hivpr_2311      QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivpr_2311      QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivpr_2311      QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivpr_2311      QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivpr_2311      QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
```

```

*****
101      .      .      .      .      150

151      .      .      .      .      198
[Truncated_Name:1]hivpr_2311  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_2311  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_2311  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_2311  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_2311  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
151      .      .      .      .      198

```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exe_file = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

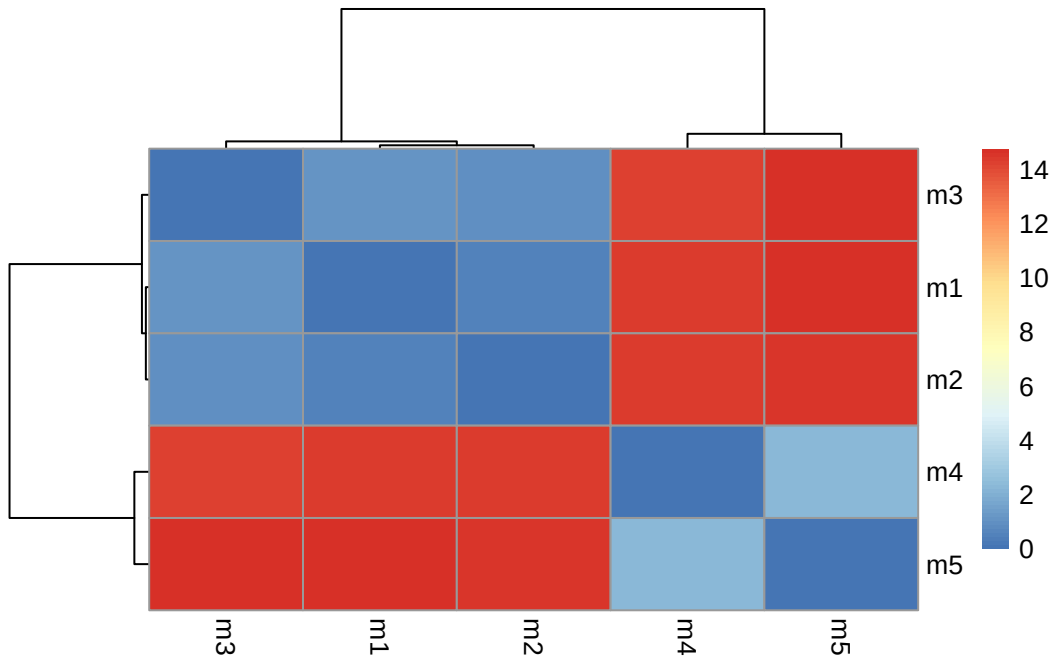
```
rd <- rmsd(pdb, fit=T)
```

Warning in rmsd(pdb, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 14.754
```

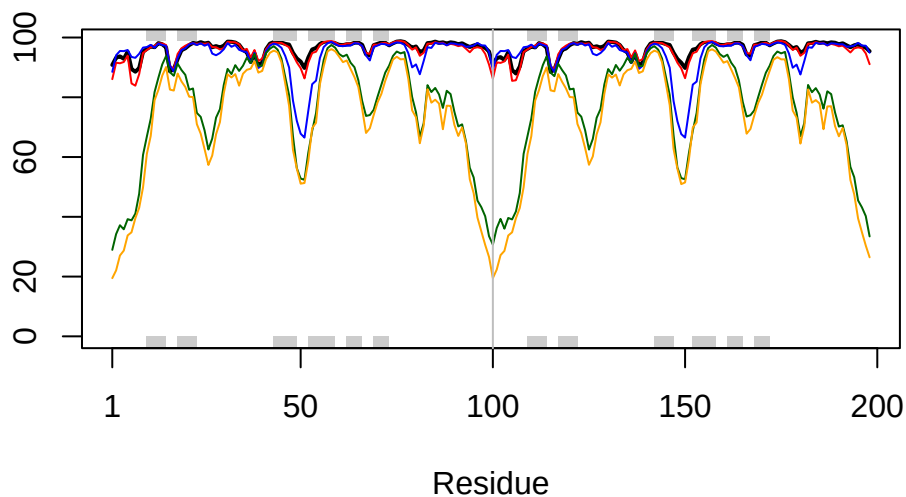
```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
core <- core.find(pdb)
```

```
core size 197 of 198 vol = 9885.821
core size 196 of 198 vol = 6896.66
core size 195 of 198 vol = 1337.842
core size 194 of 198 vol = 1040.672
core size 193 of 198 vol = 951.859
core size 192 of 198 vol = 899.085
core size 191 of 198 vol = 834.735
core size 190 of 198 vol = 771.341
core size 189 of 198 vol = 733.067
core size 188 of 198 vol = 697.282
core size 187 of 198 vol = 659.744
core size 186 of 198 vol = 625.275
core size 185 of 198 vol = 589.542
core size 184 of 198 vol = 568.255
core size 183 of 198 vol = 545.016
core size 182 of 198 vol = 512.891
core size 181 of 198 vol = 490.725
core size 180 of 198 vol = 470.267
core size 179 of 198 vol = 450.732
core size 178 of 198 vol = 434.736
```

core size 177 of 198	vol = 420.338
core size 176 of 198	vol = 406.659
core size 175 of 198	vol = 393.335
core size 174 of 198	vol = 382.396
core size 173 of 198	vol = 372.859
core size 172 of 198	vol = 356.994
core size 171 of 198	vol = 346.568
core size 170 of 198	vol = 337.447
core size 169 of 198	vol = 326.659
core size 168 of 198	vol = 314.95
core size 167 of 198	vol = 304.128
core size 166 of 198	vol = 294.553
core size 165 of 198	vol = 285.649
core size 164 of 198	vol = 278.885
core size 163 of 198	vol = 266.766
core size 162 of 198	vol = 258.995
core size 161 of 198	vol = 247.723
core size 160 of 198	vol = 239.841
core size 159 of 198	vol = 234.963
core size 158 of 198	vol = 230.062
core size 157 of 198	vol = 221.985
core size 156 of 198	vol = 215.621
core size 155 of 198	vol = 206.793
core size 154 of 198	vol = 196.984
core size 153 of 198	vol = 188.54
core size 152 of 198	vol = 182.262
core size 151 of 198	vol = 176.954
core size 150 of 198	vol = 170.712
core size 149 of 198	vol = 166.12
core size 148 of 198	vol = 159.797
core size 147 of 198	vol = 153.767
core size 146 of 198	vol = 149.092
core size 145 of 198	vol = 143.657
core size 144 of 198	vol = 137.138
core size 143 of 198	vol = 132.517
core size 142 of 198	vol = 127.231
core size 141 of 198	vol = 121.574
core size 140 of 198	vol = 116.775
core size 139 of 198	vol = 112.57
core size 138 of 198	vol = 108.17
core size 137 of 198	vol = 105.133
core size 136 of 198	vol = 101.249
core size 135 of 198	vol = 97.374

core size 134 of 198	vol = 92.974
core size 133 of 198	vol = 88.184
core size 132 of 198	vol = 84.029
core size 131 of 198	vol = 81.898
core size 130 of 198	vol = 78.019
core size 129 of 198	vol = 75.272
core size 128 of 198	vol = 73.052
core size 127 of 198	vol = 70.695
core size 126 of 198	vol = 68.975
core size 125 of 198	vol = 66.694
core size 124 of 198	vol = 64.394
core size 123 of 198	vol = 62.092
core size 122 of 198	vol = 59.045
core size 121 of 198	vol = 56.629
core size 120 of 198	vol = 54.016
core size 119 of 198	vol = 51.805
core size 118 of 198	vol = 49.652
core size 117 of 198	vol = 48.193
core size 116 of 198	vol = 46.648
core size 115 of 198	vol = 44.752
core size 114 of 198	vol = 43.292
core size 113 of 198	vol = 41.093
core size 112 of 198	vol = 39.147
core size 111 of 198	vol = 36.471
core size 110 of 198	vol = 34.117
core size 109 of 198	vol = 31.47
core size 108 of 198	vol = 29.447
core size 107 of 198	vol = 27.325
core size 106 of 198	vol = 25.822
core size 105 of 198	vol = 24.15
core size 104 of 198	vol = 22.648
core size 103 of 198	vol = 21.069
core size 102 of 198	vol = 19.953
core size 101 of 198	vol = 18.3
core size 100 of 198	vol = 15.723
core size 99 of 198	vol = 14.841
core size 98 of 198	vol = 11.646
core size 97 of 198	vol = 9.434
core size 96 of 198	vol = 7.354
core size 95 of 198	vol = 6.179
core size 94 of 198	vol = 5.665
core size 93 of 198	vol = 4.705
core size 92 of 198	vol = 3.665


```

core size 91 of 198  vol = 2.77
core size 90 of 198  vol = 2.151
core size 89 of 198  vol = 1.715
core size 88 of 198  vol = 1.15
core size 87 of 198  vol = 0.874
core size 86 of 198  vol = 0.685
core size 85 of 198  vol = 0.528
core size 84 of 198  vol = 0.37
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44

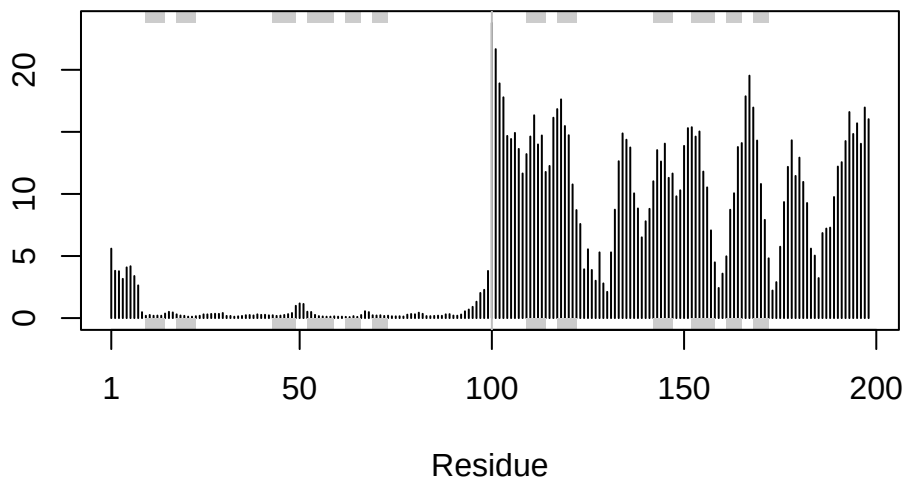
```

```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```

rf <- rmsf(xyz)
plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```



Predicted Alignment Error for domains

```
library(jsonlite)
```

```
# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt" "max_pae" "pae" "ptm" "iptm"
```

```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

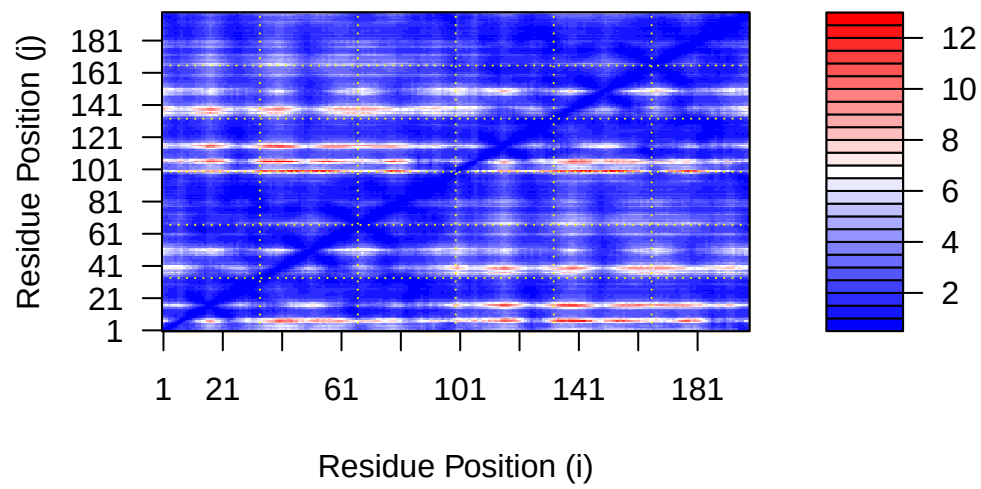
```
pae1$max_pae
```

```
[1] 12.84375
```

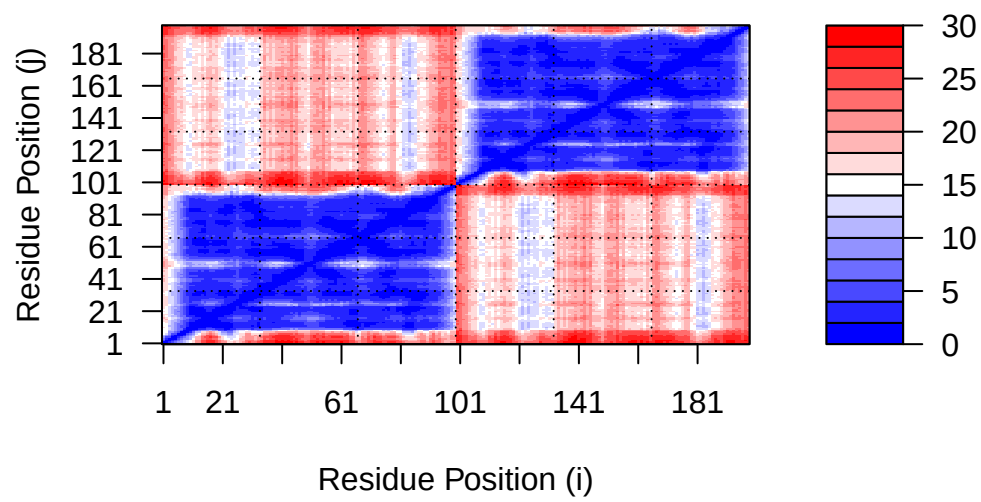
```
pae5$max_pae
```

```
[1] 29.59375
```

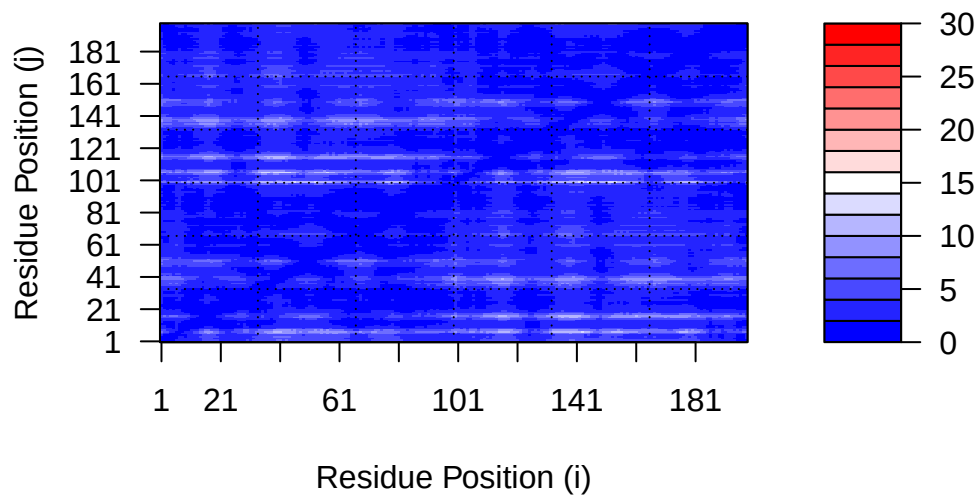
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

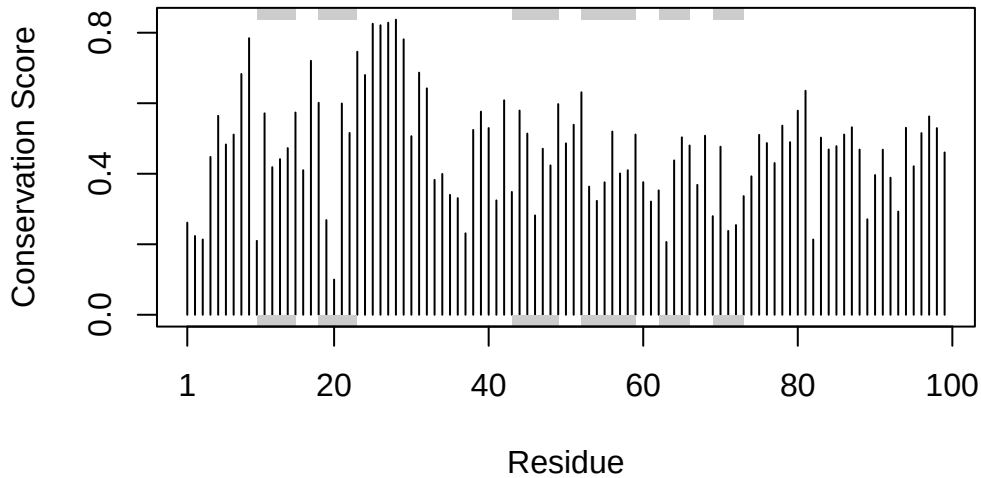
```
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```